Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Schiffer, Peter Ph.D.		
Project Title:	Experimental and Theoretical Studies of Wet Granular Media		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	University Park	**State:**	PA
Zip Code:	16802	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	03/01/2000	**End Date:**	07/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:	Agui, J. H.	**Contact Phone:**	216-433-5409
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Flight Program:			
Flight Assignment:	grant extended 1 yr		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Barabasi, Albert-Laszlo (University of Notre Dame)		
Grant/Contract No.:	NAG3-2546		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	These experimental and theoretical studies require examination of how such wetting affects the physical properties of granular materials. In order to understand the complex physics involved new methods must be developed and tested on dry media as a preliminary to studying the wet grains. Planned experimental studies include drag force experiments and avalanching experiments. This project entails studies of how small amounts of interstitial liquid affect the static and dynamic properties of granular media. Of particular interest is the development of inter-grain correlations, i.e., clumping. The physical properties of granular materials are very complex, and they include features, which are both solid-like and liquid-like. Understanding these properties is of immense practical importance since billions of dollars of industrial output relies on the processing of granular materials. The motivation for this work is to better understand the effects of		

	wetting on granular materials. This is important both from a fundamental standpoint since adhesive forces change the physics of granular materials and also from a commercial standpoint because wetting is relevant to many industrial applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 11/15/2007)